

Closure of Congenital Semicircular Dehiscence in the First 10 Years of Life

Schluss der kongenitalen Bogengangsdehiszenz im Zuge des physiologischen Kopfwachstums

Authors

Katja Döring¹, Heinrich Lanfermann¹, Athanasia Warnecke², Anja Gieseemann¹

Affiliations

- 1 Department of Diagnostic and Interventional Neuroradiology, Hannover Medical School, Hanover, Germany
- 2 Department of Otorhinolaryngology, Hannover Medical School, Hannover, Germany

Keywords

third window, CBCT, HRCT, semicircular dehiscence

received 28.8.2025

accepted after revision 14.10.2025

published online 11.11.2025

Bibliography

Rofo 2026; 198: 781–788

DOI 10.1055/a-2725-8648

ISSN 1438-9029

© 2025, Thieme. All rights reserved.

Georg Thieme Verlag KG, Oswald-Hesse-Straße 50, 70469 Stuttgart, Germany

Correspondence

Katja Döring

Department of Diagnostic and Interventional Neuroradiology, Hannover Medical School, Hanover, Germany
doering.katja@mh-hannover.de

ABSTRACT

Purpose To assess the long-term progression of semicircular canal dehiscence (SSCD), particularly with regard to size expansion or possible ossification during physical development.

Materials and Methods This retrospective study screened all patients under the age of three who underwent HRCT between 2006 and 2013 for various indications, e.g. diagnosis of sensorineural hearing loss (SNHL), acute mastoiditis, or to evaluate for cochlear implantation. The study population was also screened for the presence of follow-up imaging. Only patients with imaging data available after at least two years or more were included in the study. Demographic data, clinical symptoms, and imaging findings were analyzed using descriptive statistics. The imaging findings of semicircular canal dehiscence were categorized by location and severity, and the long-term development was evaluated. More than 1500 HRCTs of the temporal bone taken in children ages 0 to

10 years old from 2006 to 2025 were additionally examined for the presence of SCC dehiscence.

Results Of the 586 HRCT scans performed on young patients under three years of age, 93 examinations for which follow-up examinations were available two or more years later could be recorded. These were distributed among 32 children with an average age of 11.3 ± 7.4 months (range 3–24 months). At the time of initial imaging (baseline examination), 46 SCCDs of the superior or posterior semicircular canal were identified. A total of 36 ears were affected. In the subsequent HRCT examination, conducted 2–13 years later, the extent of bone coverage was consistently quantifiable. The mean thickness of the bone over the site of the previous dehiscence was 1 mm (range 0.2–3.3 mm).

Conclusion In the vast majority of cases, the SCC dehiscence observed in studies of the temporal bones of young children will be covered by normal bone growth by puberty. This suggests that SCC dehiscence does not persist from birth into adulthood, and instead it supports the theory of an acquired pathogenesis of SCC dehiscence.

Key Points

- In children, SSCD is most likely to be completely covered by skull growth.
- Clinically significant SSCD in adults is most likely to be acquired (not present since childhood).
- There is a high prevalence of SSCD between the ages of 0–6 and 6–12 months.
- SSCD occurs more frequently than PSCD overall.
- The prevalence of SSCD decreases as age increases.

Citation Format

- Döring K, Lanfermann H, Warnecke A et al. Closure of Congenital Semicircular Dehiscence in the First 10 Years of Life. Rofo 2026; 198: 781–788

ZUSAMMENFASSUNG

Ziel Bewertung des langfristigen Verlaufs einer Dehiszenz des Bogengangs (SCD), insbesondere hinsichtlich Ausdehnung oder möglicher Ossifikation während der körperlichen Entwicklung.

Materialien und Methoden In dieser retrospektiven Studie wurden alle Patienten im Alter von unter drei Jahren untersucht, die zwischen 2006 und 2013 aus verschiedenen Grün-

den einer HRCT unterzogen wurden, z. B. zur Diagnose einer sensorineuralen Hörminderung (SNHL), einer akuten Mastoiditis und/oder zur Beurteilung einer Cochlea-Implantation. Die Studienpopulation wurde auf das Vorliegen von Follow-up-Bildgebungen gescreent. Eingeschlossen wurden lediglich die Patienten, bei denen ein Follow-up von mindestens zwei Jahren vorlag. Demografische Daten, klinische Symptome und Bildgebungsbefunde wurden mittels deskriptiver Statistik analysiert. Die Bildgebungsbefunde einer Dehiszenz des Bogengangs wurden nach Lage und Schweregrad kategorisiert und die Langzeitentwicklung bewertet. Zusätzlich wurden mehr als 1500 HRCTs des Schläfenbeins, die im Zeitraum von 2006 und 2025 bei Kindern im Alter von 0 bis 10 Jahren erfolgten, auf das Vorliegen einer SCD untersucht.

Ergebnisse Von 586 HRCT-Untersuchungen bei jungen Patienten unter drei Jahren konnten 93 Untersuchungen erfasst werden, für die zwei oder mehr Jahre später Follow-up-Bildgebungen vorlagen. Diese verteilten sich auf 32 Kinder mit einem Durchschnittsalter von $11,3 \pm 7,4$ Monaten (Bereich 3–24 Monate). Bei der ersten Bildgebung (Ausgangsuntersuchung) wurden 46 SCDs des oberen oder hinteren Bogengangs festgestellt. Insgesamt waren 36 Ohren betrof-

fen. Bei der Nachuntersuchung, die 2–13 Jahre später stattfand, war die Knochenbedeckung im HRCT-Scan immer messbar. Die durchschnittliche Knochendicke über der ehemaligen Dehiszenz betrug 1 mm (Bereich 0,2–3,3 mm).

Schlussfolgerung Die überwiegende Mehrheit der in Studien an den Schläfenbeinen von Kleinkindern beobachteten Fälle von SCD wird bis zur Pubertät durch normales Knochenwachstum abgedeckt sein. Dies deutet darauf hin, dass die SCCD nicht von der Geburt bis ins Erwachsenenalter bestehen bleibt, was stattdessen die Theorie einer erworbenen Pathogenese der SCCD stützt.

Kernaussagen

- Eine SCCD bei Kindern wird höchstwahrscheinlich durch das Schädelwachstum vollständig gedeckt.
- Eine klinisch signifikante SCCD beim Erwachsenen ist höchstwahrscheinlich erworben (nicht ab Kindheit).
- Es gibt eine hohe Prävalenz von SCCD im Alter zwischen 0 und 6 sowie zwischen 6 und 12 Monaten.
- SSCD tritt insgesamt häufiger auf als PSCD.
- Die Prävalenz von SCCD nimmt mit zunehmendem Alter ab.

Abbreviations

CBCT	cone beam computed tomography
HRCT	high-resolution computed tomography
SNHL	sensorineural hearing loss
SCC	semicircular canals
SCCD	semicircular canals dehiscence
SSCD	superior semicircular canal dehiscence
PSCD	posterior semicircular canal dehiscence

Introduction

Acute vertigo is one of the most common symptoms encountered by ENT doctors in emergency rooms, and it affects around 20 to 30% of the total population [1, 2, 3, 4, 5]. The lifetime prevalence of vertigo with vestibular origin is 7.4%. Women are affected more often than men. In over 88% of cases, vertigo is recurrent [1, 2, 3, 4, 5]. The most common peripheral causes of vertigo are benign paroxysmal positional vertigo, labyrinthitis, vestibular neuronitis, and Ménière's disease. Less common causes include vestibular schwannomas, growing intrameatally, and vestibular manifestations of systemic diseases [5, 6].

Even less well known is the symptom complex of the “third window,” which was first described by Minor et al. in 1998 [6]. The complex of symptoms described includes noise (classic Tullio phenomenon) and pressure-induced vertigo (e. g. when sneezing, coughing, pressing, or lifting heavy loads), as well as dull ear

pressure, hearing loss, and autophony [7]. These symptoms are caused by a “third window” (in addition to the oval and round windows), which is a segmental dehiscence (a missing bone covering) of one of the three semicircular canals in the vestibular system.

The “third window” is typically located in the upper semicircular canal, and less commonly in the posterior or lateral semicircular canal. Based on cadaver studies, the incidence of superior semicircular canal dehiscence (SSCD) is 0.3% [8, 9]. This contrasts with the 9% prevalence confirmed or diagnosed by radiologists [9]. It can be assumed that this discrepancy between radiological findings and autopsy studies is due to the low specificity of imaging, leading to overdiagnosis. The condition described by Zhou et al. as a major otological mimicker is considered to be a disease of adulthood [10]. The peak incidence is in the fifth decade of life, with men being affected slightly more frequently [10]. However, some studies have also reported a high incidence in pediatric populations [11, 12, 13]. Both Chen et al. [13] and Meiklejohn et al. [7] investigated the prevalence and development of semicircular canal dehiscence (SCCD) in pediatric populations using long-term studies. They confirmed that overall prevalence is high in infants, but decreases during the first 10 years of life [7, 11, 12, 13].

The present study aims to determine whether a dehiscence in one of the three arches in infants represents incomplete ossification or whether it already corresponds to permanent dehiscence in adulthood.

Material and Methods

Our ethics committee approved this retrospective study and waived the requirement for informed patient consent.

Patient selection

In a tertiary referral center, all patients were referred from the Department of Otolaryngology for assessment of sensorineural hearing loss (SNHL) or for evaluation for cochlear implantation. This usually involved imaging, including a cone beam or high-resolution computed tomography (CBCT or HRCT) of the temporal bone and an MRI scan of the temporal bone. The focused search included the pediatric collective (HRCTs only): All children under three years of age examined between 2006 and 2013 were included, provided they had had a follow-up CT scan more than two years later, regardless of age or gender. From this group, only patients with a detectable semicircular canal dehiscence (SCCD) were included in the present study. Demographic data and additional clinical information were added. More than 1500 HRCTs of the temporal bone acquired between 2006 and 2025 in children ages 0 to 10 years old were additionally examined for the presence of SCC dehiscence and its severity in order to determine prevalence rates for defined age groups.

Imaging of the temporal bone

All pediatric patients underwent HRCT of the temporal bone. Different types of HRCT scanners (HiSpeed Advantage RP, HiSpeed Advantage, and LightSpeed16; all GE, Milwaukee, WI, USA) were used over the long period of our study and slice thickness ranged from 0.625 to 1 mm. The acquired images were uploaded to our current PAC system (GE, Milwaukee, WI, USA) and VISAGE Imaging (Visage 7, Pro Medicus Limited, Berlin, Germany). CBCTs were not considered, because children under the age of three are routinely examined using HRCT at our clinic.

Radiological assessment of SCCD

Two neuro-/radiologists reviewed all imaging studies in order to determine the extent, degree, and location of dehiscent defects in the SCC. The dehiscence was categorized as either questionable thin coverage (grade 3) or clear dehiscence (grade 4). In each case, bone thickness was assessed and measured at the corresponding site of the initial dehiscence in the follow-up CT scan. These measurements were taken from reformatted slices in the plane of the SCC and perpendicular to it. Records of pediatric patients, who underwent routine computed tomography (HRCT) of the temporal bone as part of an investigation by various otological clinics, were evaluated with regard to the indication for CT and the presence of malformations and syndromes. The layer thickness was also recorded in each case. The imaging findings were documented by consensus.

Statistical analysis

Statistical analysis was performed using Spss, Version 29.0. Descriptive statistics were used to analyze the study population, the type, and quality of neuroimaging, imaging findings, patient sta-

tus, and clinical symptoms. Continuous variables, such as patient age, are expressed as means, medians, and ranges. Categorical variables and qualitative parameters, such as patient sex and age. Means and standard deviation (SD) were calculated for metric variables (i. e. age).

Results

A retrospective study screened 586 HRCTs acquired from the youngest patients (0 to 3 years old) between 2006 and 2013 for the presence of follow-up imaging after at least two years or more. In most cases ($n = 7$), the second imaging examination was performed within four years of the initial examination. Twenty-two patients had a follow-up period of more than five years.

Baseline characteristics of the study population

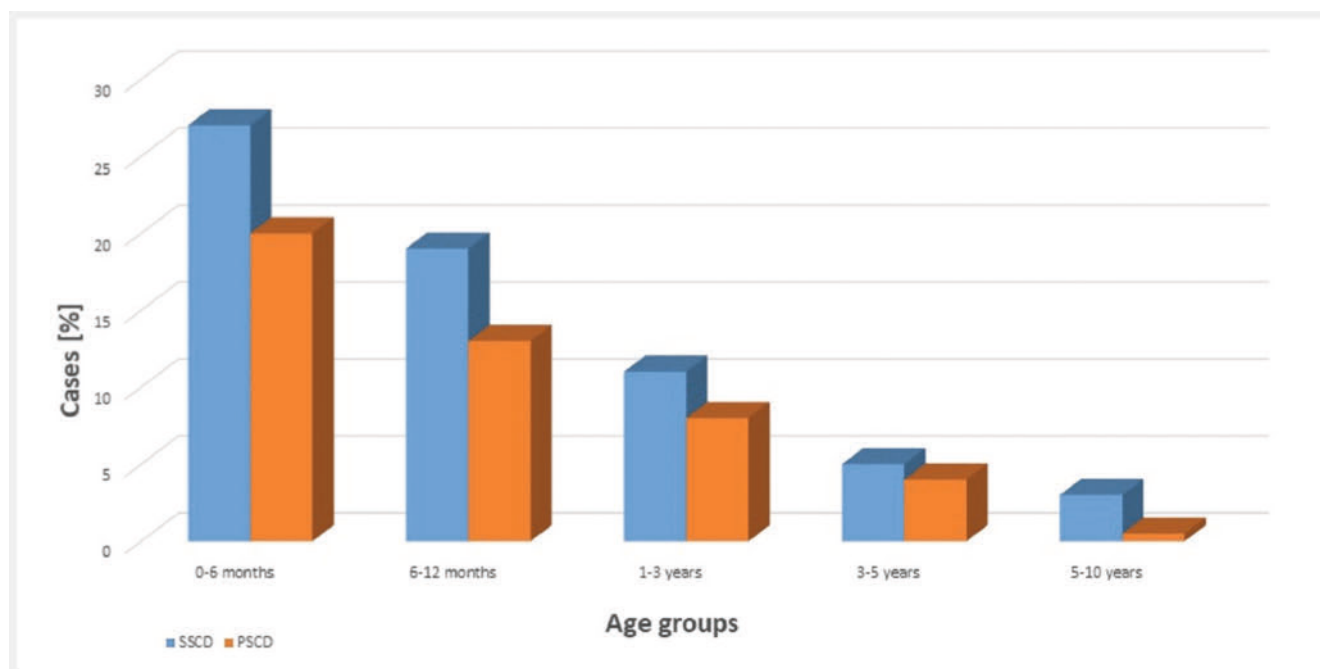
The 32 patients included in the study were between the ages of 2 to 24 months at the time of the baseline examination. The average age was 11.3 ± 7.4 months. The average follow-up period was 7.6 years, ranging from 2 to 13 years. Thirty-six ears (56.25%) were affected by SCCD. The most common finding was a dehiscence of the superior semicircular canal (SCCD) ($n = 25$, 69.4%).

Continued dehiscence was confirmed in follow-up imaging in two cases, with no change in severity or location. The affected patients were four and six years old at the time of follow-up imaging. Dehiscence could no longer be detected in the other 30 patients. The previously affected areas of the archways showed an average coverage of 1 mm (range 0.2–3.3 mm). The children with the most extensive coverage, at 2.7 and 3.3 mm, respectively, were ages 11 and 12 at the time of follow-up (► Fig. 1, ► Fig. 2, ► Fig. 3, ► Fig. 4). There was no positive correlation between age and the thickness of bone coverage of the dehiscence.

Prevalence of SCCD

Between 2006 and 2025, a total of 1,647 pediatric HRCTs (3,394 ears) were re-examined for dehiscence in one of the semicircular canals. The patients were divided into the following age groups: 0–6 months, 6–12 months, 1–3 years, 3–5 years, and 5–10 years. The smallest group was the 0–6 months age group, with $n = 165$ cases, which is the limiting factor here. This is due to the cautious use of ionizing radiation and child development. Special examinations and treatment decisions for cochlear implants in cases of SNHL are routinely made from the age of six months onwards, which is why this group is significantly larger.

Accordingly, this classification alone led to uneven group distribution and distorted data, which was taken into account when interpreting the results. However, as ► Fig. 1 shows, SSCD is much more prevalent than PSCD across all age groups. PSCD is extremely rare, particularly in older age groups. In percentage terms, the 0–6 months age group is the most severely affected by SSCD.



► **Fig. 1** Distribution of cases of SSCD and PSCD among the different age groups over the entire period from 2006 to 2025. (SSCD: superior semicircular canal dehiscence; PSCD: posterior semicircular canal dehiscence).

Discussion

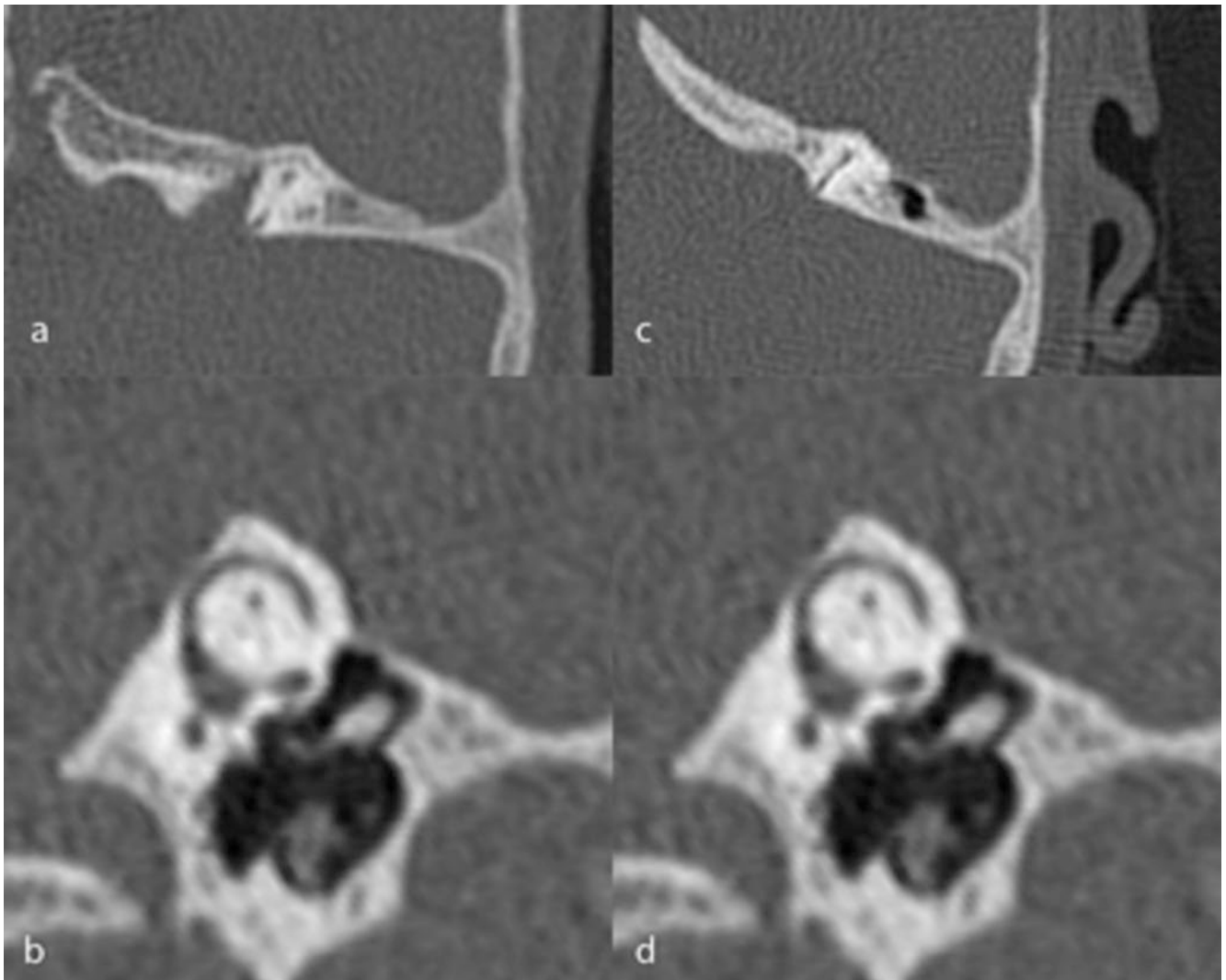
On average, sufficient bone coverage of the pre-existing arch canal dehiscence was achieved over a period of 7.6 years. In two cases (6.25%), the dehiscence remained visible; however, neither showed any dynamic changes in terms of their dimensions or position. The thickness of the bone coverage varied considerably and showed no significant correlation with the patients' age.

This study confirms the findings of Meiklejohn et al. [7], who conducted a large retrospective study in 2015 examining 228 HRCTs and 58 temporal bone samples from children ages 0 to 7 years for the presence of semicircular canal dehiscence. The histopathological data obtained from this study confirmed that semicircular canal dehiscence is not only a common radiological finding in children. The study group led by Meiklejohn et al. determined a prevalence of SSCD of 11.9%, 4.9%, 2.8% and 0%, and that of PSCD in children under 6 months, 6 to 11 months, 12 to 35 months, and 3 to 7 years at 16.7%, 2.4%, 1.4% and 0% [7].

The hypothesis proposed by Meiklejohn et al. that the bone thickness of the arch canals increases in volume with age [7] is confirmed by our study results: In 93.75% of cases, the dehiscence was covered with bone during the follow-up period. Only in two patients, who were four and six years old at the time of subsequent imaging, did the dehiscence persist. Accordingly, it can be concluded that in children, arch dehiscence observed on HRCT corresponds to incomplete bone growth. The absence of accompanying symptoms, as noted by Chen et al. [13], supports the hypothesis that such findings are rarely clinically significant. Consequently, observed SSCD should be given special attention as a radiological pitfall in young patients. Last but not least, the extent to which insufficient HRCT resolution is effective in this

context should be discussed. If the bony structures of the arches are too thin, partial volume effects can make the arches appear open. Accordingly, caution is advised when diagnosing SSCD in children. To underscore the importance of high-resolution imaging using the thinnest possible slice thickness, Belden et al. 2003 initiated a study in which 36 cases were compared based on the selected slice thickness (1 mm vs. 0.5 mm), thus refuting all diagnoses of SSCD initially made on the basis of the 1 mm slice thickness [14].

However, the case report by Zhou et al. [15] shows that, in individual cases, SSCD might very well be a structural cause of auditory and vestibular symptoms in children: The report describes the case of a 12-year-old patient who presented with symptoms of vertigo and balance disorders [15]. Right-sided SSCD had already been diagnosed at the age of seven on the basis of a CCT as part of cerebral palsy diagnosed in early childhood, and it persisted until the age of 12 years [15]. An audiological examination confirmed conductive hearing loss on the same side, confirming the diagnosis of third-window syndrome after other possible causes had been ruled out [15]. In 1998, Minor et al. first reported on eight patients with SSCD who suffered from oscillopsia and vertigo symptoms triggered by pressure or noise [6]. Clinical studies on SSCD have shown that patients may also experience auditory symptoms such as conductive hearing loss and tinnitus with or without vestibular symptoms [1, 2, 3, 4, 5, 6]. The "third window effect" theory was proposed to explain these clinical manifestations [1, 2, 3, 4, 5, 6]. The result of the open semicircular canal is that a third movable window diverts sound energy away from the cochlea, leading to pseudo-conductive hearing loss [1, 2, 3, 4, 5, 6]. Furthermore, it has been shown that the acoustic energy diverted into the vestibule by the dehiscence cau-



► **Fig. 2** Cone beam images of the semicircular canal (SCC) of a 2-year-old and 5-month-old girl in an (a) axial plane and (b) the plane of the SSC (Pöschl plane), showing a dehiscence in the superior semicircular canal (SSCD). Both at two different points in time. The second point in time at the age of 6 years and 9 months, in (c) axial and (d) Pöschl plane (d). Images c and d discreetly show the growth of the temporal bone covering the SCC.

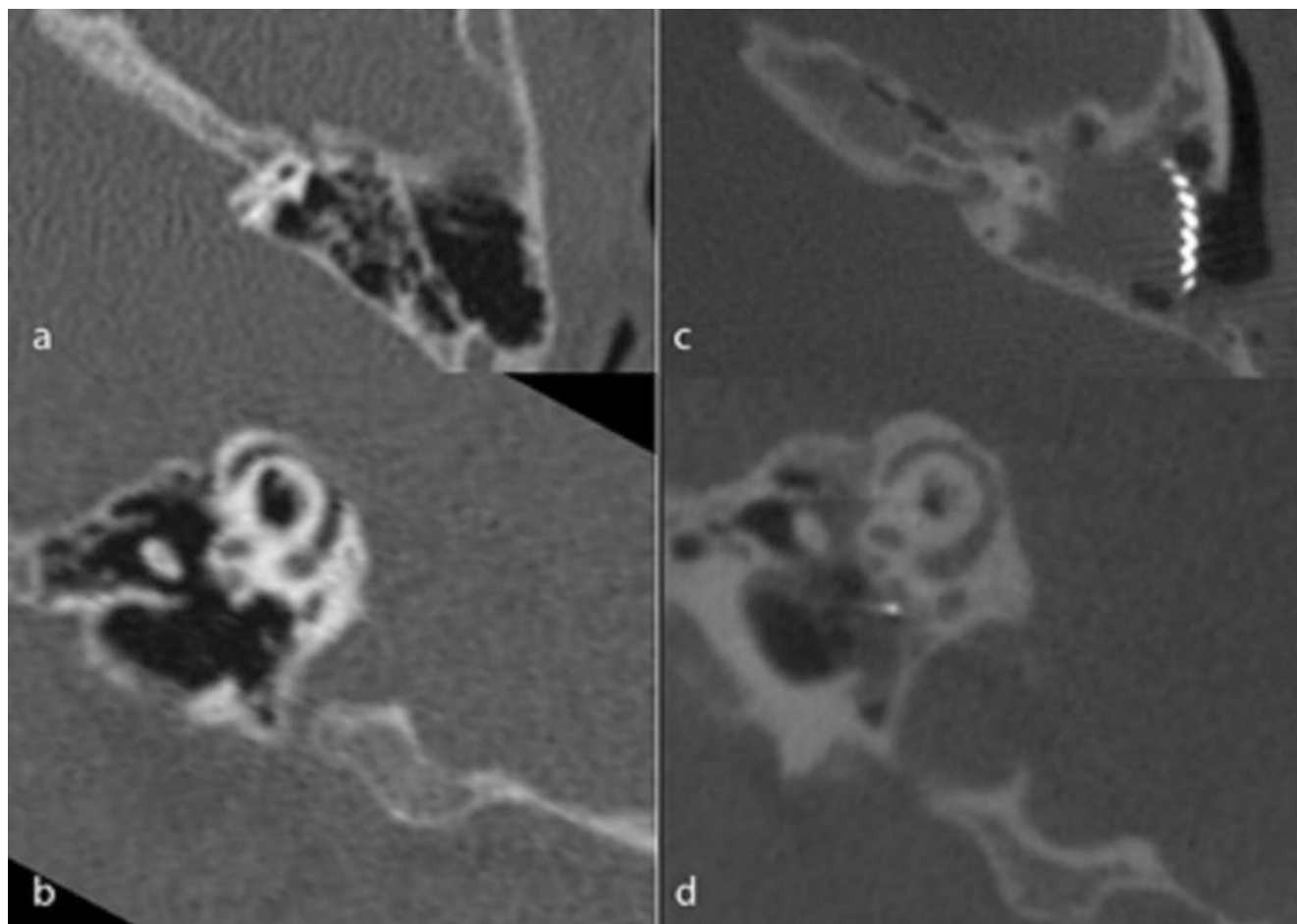
ses increased fluid flow in the inner ear and induces excitation of the cupula, leading to symptoms of vestibular dysfunction [1, 2, 3, 4, 5, 6].

Notably, there are few publications on the occurrence of SSCD in early adulthood. Most cases are diagnosed in patients between the ages of 50 and 60 years [12]. This raises the question of whether SSCD is actually a congenital or developmental defect that only manifests itself later in life. A variety of research studies, including both histological and imaging examinations of the temporal bone, support the hypothesis that SSCD may be based on a developmental disorder [15, 16]. Carey et al. focused, in particular, on bone samples from infants [8]. At birth, the samples showed uniformly thin bone above the superior canal, which gradually thickened until the age of three. In temporal bone samples from adults, he found SSCD in approximately 0.5% of cases [8]. He concluded that this anomaly could be the result of impaired postnatal bone development [8]. He also discussed the hypothesis that areas with thin bone above the superior canal are

susceptible to traumatic injury and could therefore also be of post-traumatic origin [8].

Crovetto et al. considered SSCDs in the broader context of aging and menopause, i. e. developmental phases associated with increased bone loss [17]. In a prospective observational study involving 582 ears from 312 patients, a 10% (0.1 mm) decrease in bone thickness above the SSC was observed [17]. The average bone thickness above the SSC was 1.14 mm (SD 0.52) in subjects under 45 years of age and 1.02 mm (SD 0.45; $p=0.006$) in subjects over 45 years of age [17]. Crovetto et al. concluded that there is a positive correlation between osteopenia of the roof of the superior semicircular canal and age, with this effect appearing to be more pronounced in postmenopausal women [17].

Given the high prevalence of SSCD syndrome in older people, Minor et al. [6] and Berkiten et al. [18] provided another hypothetical approach based on pressure-induced wear (due to axial force exerted by the temporal lobe and cerebrospinal fluid) of the bony covering of the superior semicircular canal [6]. This hypothesis as-



► **Fig. 3** Cone beam images of the semicircular canal (SCC) of an eight-month-old boy in axial (a) and Pöschl plane (b), showing a dehiscence in the superior semicircular canal (SSCD). The next two images show the same child aged 4 years and 9 months, again in axial (c) and Pöschl (d) plane, illustrating bone growth. A CI operation took place in between.

sumes that SSCD occurs more frequently in older people, as do the stages prior to dehiscence. Conversely, the maximum thickness would be found in younger subjects. Accordingly, he and his team examined a study population for the presence of SSCD in order to prove a correlation between age and SSCD, which he succeeded in doing [6]. Nadgir et al. [12] also confirmed that the prevalence of SSCD per 10-year age group (in 608 scans) increased significantly by 93%. In his study group, he found a 33% prevalence of SSCD in the 81–100 age group. Consequently, age and population aging appear to be factors that increase the prevalence of SSCD.

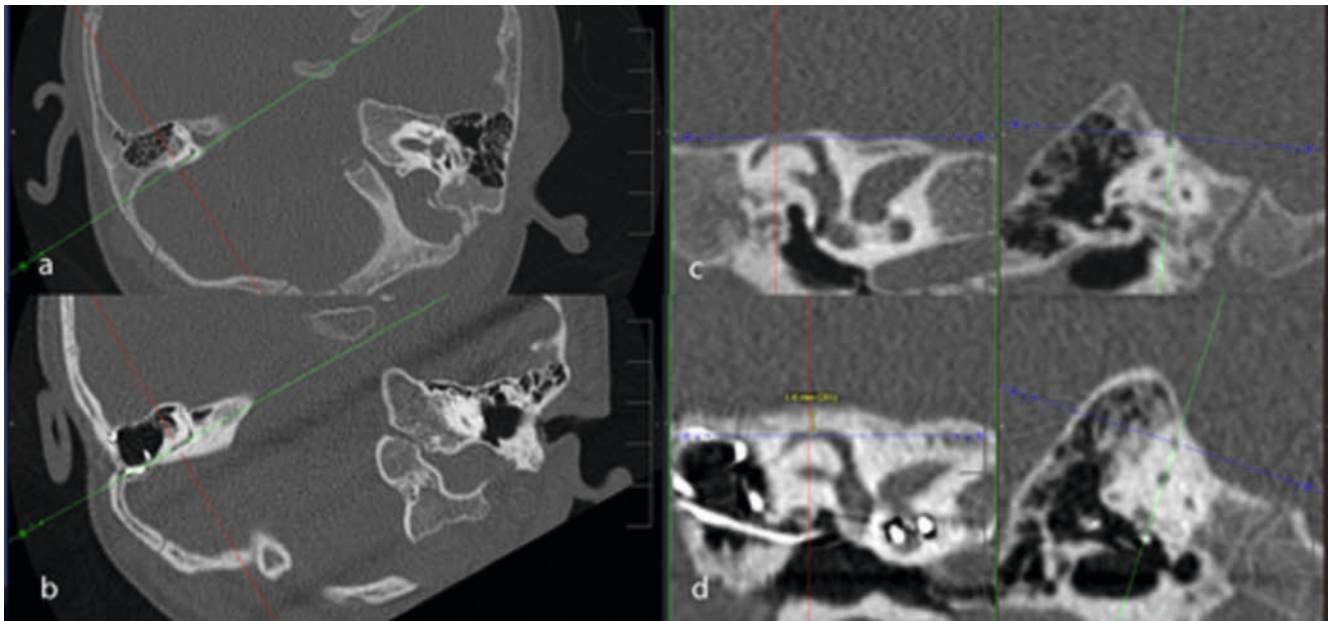
According to Waldeck et al. [19], who published a new SSCD classification in 2022, there are no significant differences in SSCD incidence rates between women and men. In line with existing literature, subgroup analysis confirmed that SSCD type 2 (defined as the highest point of the superior semicircular canal circumference) occurs significantly more frequently in men, particularly in the 50–74 and >75 age groups. Waldeck et al. [19] also emphasize the importance of understanding SSCD syndrome. By introducing a new classification system for SSCD, they highlight the pitfalls in diag-

nosing SSCD. Using the proposed reconstruction scheme significantly reduces the risk of overdiagnosis when using CT [19].

To address the recommendation to perform a high-resolution CT (HRCT) evaluation in cases of questionable SSCD findings on MRI, Rabiei et al. [20] screened over 5000 MRI examinations to recommend further clarification of possible SSCD using CT. Of the remaining 101 patients, only 32% actually had SSCD, and one of these patients suffered from dehiscence syndrome. Accordingly, the study group concluded that recommending further clarification of questionable SSCD findings on MRI has negligible clinical value.

Limitations

In addition to the retrospective, monocentric study design, the long observation period is also a limitation and necessitated the use of different CT scanners. This has limited the ability to conduct immediate one-to-one assessments, particularly with regard to critical details such as density in small areas. In addition, artifacts caused by the cochlear implant (CI) made assessment more challenging.



► **Fig. 4** Multiplanar reformation of temporal bone CTs at two different points in time. Upper row: segmental dehiscence of the superior semicircular canal (SSCD) on the right. Lower row: there is now clear bony coverage of the superior semicircular canal on the right. The previous dehiscence is no longer visible. Adjunct finding: cochlear implant inserted in the meantime on the right.

Conclusion

The vast majority of cases of SCC dehiscence observed in studies of the temporal bones of young children will be covered by normal bone growth by puberty. This suggests that SCC dehiscence does not persist from birth into adulthood, and instead it supports the theory of an acquired pathogenesis of SCC dehiscence.

Conflict of Interest

The authors declare that they have no conflict of interest.

References

- [1] Neuhauser HK, von Brevern M, Radtke A et al. Epidemiology of vestibular vertigo: A neurotologic survey of the general population. *Neurology* 2005; 65: 898–904. doi:10.1212/01.wnl.0000175987.59991.3d
- [2] Ibekwe TS, Fasunla JA, Ibekwe PU et al. Migraine and Meniere's disease: Two different phenomena with frequently observed concomitant occurrences. *J Natl Med Assoc* 2008; 100: 334–338
- [3] Sajjadi H, Paparella MM. Meniere's disease. *Lancet* 2008; 372: 406–414. doi:10.1016/S0140-6736(08)61161-7
- [4] Furman JM, Cass SP. Benign paroxysmal positional vertigo. *N Engl J Med* 1999; 341: 1590–1596. doi:10.1056/NEJM199911183412107
- [5] Kontorinis G, Tyagi A, Crowther JA. Recurrent vertigo associated with headaches. *BMJ* 2018; 363: k1807. doi:10.1136/bmj.k1807
- [6] Minor LB, Solomon D, Zinreich JS et al. Sound- and/or pressure-induced vertigo due to bone dehiscence of the superior semicircular canal. *Arch Otolaryngol Head Neck Surg* 1998; 124: 249–245. doi:10.1001/archotol.124.3.249
- [7] Meiklejohn DA, Corrales CE, Boldt BM et al. Pediatric semicircular canal dehiscence. *Otol Neurotol* 2015; 36: 1383–1389. doi:10.1097/MAO.0000000000000811
- [8] Carey JP, Minor LB, Nager GT. Dehiscence or thinning of bone overlying the superior semicircular canal in a temporal bone survey. *Arch Otolaryngol Head Neck Surg* 2000; 126: 137–147. doi:10.1001/archotol.126.2.137
- [9] Berning AW, Arani K, Branstetter BF 4th. Prevalence of Superior Semicircular Canal Dehiscence on High-Resolution CT Imaging in Patients without Vestibular or Auditory Abnormalities. *AJNR Am J Neuroradiol* 2019; 40: 709–712. doi:10.3174/ajnr.A5999
- [10] Zhou G, Gopen Q, Poe DS. Clinical and diagnostic characterization of canal dehiscence syndrome: a great otologic mimicker. *Otol Neurotol* 2007; 28: 920–926. doi:10.1097/MAO.0b013e31814b25f2
- [11] Schutt CA, Neubauer P, Samy RN et al. The correlation between obesity, obstructive sleep apnea, and superior semicircular canal dehiscence: A new explanation for an increasingly common problem. *Otol Neurotol* 2015; 36: 551–554. doi:10.1097/MAO.0000000000000555
- [12] Nadgir RN, Ozonoff A, Devaiah AK et al. Superior semicircular canal dehiscence: Congenital or acquired condition? *AJNR Am J Neuroradiol* 2011; 32: 947–949. doi:10.3174/ajnr.A2437
- [13] Chen EY, Paladin A, Phillips G et al. Semicircular canal dehiscence in the pediatric population. *International journal of pediatric otorhinolaryngology* 2009; 73: 321–327. doi:10.1016/j.ijporl.2008.10.027
- [14] Belden CJ, Weg N, Minor LB et al. CT evaluation of bone dehiscence of the superior semicircular canal as a cause of sound- and/or pressure-induced vertigo. *Radiology* 2003; 226: 337–343. doi:10.1148/radiol.2262010897
- [15] Zhou G, Ohlms L, Liberman J et al. Superior semicircular canal dehiscence in a young child: Implication of developmental defect. *Int J Pediatr Otorhinolaryngol* 2007; 71: 1925–1928. doi:10.1016/j.ijporl.2007.08.009
- [16] Crovetto M, Whyte J, Rodriguez OM et al. Anatomic-radiological study of the Superior Semicircular Canal Dehiscence Radiological considerations of Superior and Posterior Semicircular Canals. *Eur J Radiol* 2010; 76: 167–172. doi:10.1016/j.ejrad.2009.05.038
- [17] Crovetto MA, Whyte J, Rodriguez OM et al. Influence of aging and menopause in the origin of the superior semicircular canal dehiscence. *Otol Neurotol* 2012; 33: 681–684. doi:10.1097/MAO.0b013e31824f9969

- [18] Berkiten G, Gürbüz D, Akan O et al. Dehiscence or thinning of bone overlying the superior semicircular canal in idiopathic intracranial hypertension. *Eur Arch Otorhinolaryngol* 2022; 279: 2899–2904. doi:10.1007/s00405-021-07020-z
- [19] Waldeck S, Lanfermann H, von Falck C et al. New classification of superior semicircular canal dehiscence in HRCT. *PLoS One* 2022; 17: e0262758. doi:10.1371/journal.pone.0262758
- [20] Rabiei P, Kim J, Satani AA et al. Outcomes of Radiologist Recommendations for Temporal Bone CT to Assess Superior Semicircular Canal Dehiscence on Temporal Bone MRI. *AJNR Am J Neuroradiol* 2025; 46: 1683–1687. doi:10.3174/ajnr.A8691